THE UNIVERSITY OF HONG KONG

BACHELOR OF ENGINEERING: LEVEL I EXAMINATION

DEPARTMENT OF CIVIL ENGINEERING

CIVL1105: Environmental Engineering

Dec 22, 2016 TIME: 9:30 am – 12:30 pm

I. Answer ALL questions. Section A carries 30 marks, and Section B carries 70 marks.

II. Atomic weights: H 1; C 12; N 14; O 16; F 19; Na 23; Mg 24.3; Al 27; P 31; S 32; Cl 35.5; K 39; Ca 40; Cr 52; Mn 55; Fe 55.8; Ni 58.7; Cu 63.5; Zn 65.4; I 127.

III. Properties of water:

Density = 1000 kg/m^3
Viscosity (15 °C) = 1.1404 mPa·s
Specific heat = 4.18 kJ/(kg.°C)
Heat of vaporization (15 °C) = 2465 kJ/kg
Heat of fusion = 333 kJ/kg

Use of Calculators:

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

Section A. (10 multiple choice questions with 3 marks each)

1. The alkalinity (ALK) of a water solution can be increased by the addition of chemicals. Which of the following chemicals will increase ALK? (a) $NaHCO_3$, (b) HCl, (c) CO_2, (d) $FeCl_3$, (e) NaCl, or (f) none of the above.
2. Which of the following chemicals is NOT used in the lime-soda method for hardness removal? (a) $Ca(OH)_2$, (b) Na_2CO_3, (c) CO_2, (d) NaOH, or (e) none of the above.
3. In the activated sludge process, the biomass is typically represented by (a) TDS, (b) TSS, (c) TVS, (d) TFS, (e) FSS, (f) VSS, (g) FDS, or (h) none of the above.
4. Which of the following mechanisms is NOT important in coagulation process? (a) Charge neutralization, (b) entrapment, (c) stratification, (d) inter-particle bridging, or (e) none of the above.
5. Which process is suitable for SO_2 removal from power plant flue gas? (a) Fuel cleaning, (b) three-way catalytic convertor, (c) baghouse, (d) selective catalytic reduction, (e) electrostatic precipitator, or (f) none of the above.
6. Which disinfectant below is the most effective against Cryptosporidium? (a) Chlorine dioxide, (b) chlorine, (c) carbon monoxide, (d) chloramine, or (e) ozone.
7. Which of the following compounds is a secondary air pollutant? (a) CO_2, (b) CO, (c) O_2, (d) O_3, (e) NO_x, (f) SO_x, or (g) none of the above.
8. Which biological treatment belongs to suspended aerobic treatment? (a) Sludge digestion, (b) aerated lagoon, (c) trickling filter, (d) rotating biological contactor, or (e) none of the above.
9. The F/M ratio is an important parameter in biological wastewater treatment. For the calculation of the F/M ratio, which of the following quantity is NOT needed? (a) BOD, (b) biomass concentration, (c) nitrogen concentration, (d) substrate, (e) wastewater flow rate, or (f) aeration tank volume.
10. Ocean acidification is caused by (a) CO_2, (b) NO, (c) NO_2, (d) SO_2, (e) SO_3, or (f) none of the above.

Section B (Three questions with unequal marks)

11. A lake water contains 0.05 mg/L P and 0.2 mg/L N. It is worried that eutrophication may take place in the lake.

    (a) Briefly explain the phenomenon of eutrophication. What are the typical characteristics of eutrophic lakes?

    (12 marks)

    (b) An average microalgae cell may be represented by the "chemical formula" $C_{106}H_{263}O_{110}N_{16}P$. Determine the N/P mass ratio in microalgae cells. Which element is the limiting nutrient for the lake?

    (6 marks)

    (c) What action(s) would you suggest in order to improve the water quality of the lake?

    (6 marks)

12. (a) A wastewater contains 100 mg/L C_6H_7N. Determine its carbonaceous oxygen demand and total theoretical oxygen demand.

(8 marks)

(b) A wastewater has a BOD_5 of 300 mg/L at 25°C. The rate constant at 25°C is 0.25 day^{-1}. Determine the BOD_5 at 18°C. You can assume a temperature correction factor of 1.047 in your calculation(s).

(8 marks)

(c) An untreated wastewater is accidentally discharged to a nearby river. Explain what will happen to the dissolved oxygen (DO) in the river. Sketch the DO concentration as a function of distance downstream to the wastewater discharge point.

(10 marks)

13. (a) The Monod equation is given in the form of $\mu = \mu_{max}S/(K_s + S)$. Based on this equation, explain how the microbial growth depends on the availability of substrate.

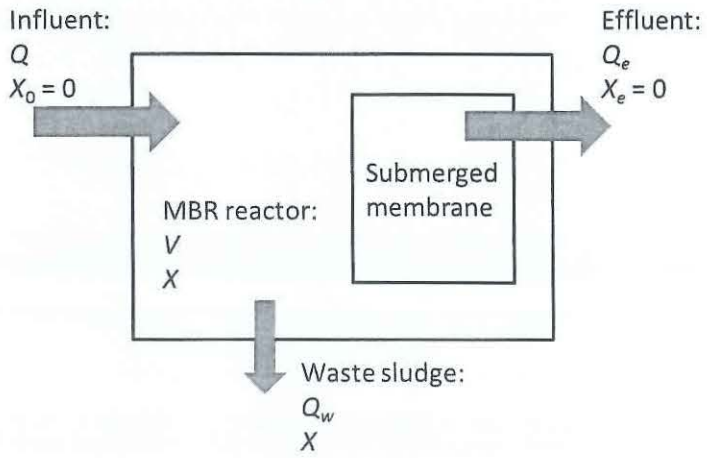
(6 marks)

(b) Membrane bioreactors (MBRs) are widely used for wastewater treatment. What are the common type(s) of membranes used in MBRs? What is the main purpose of using membranes?

(4 marks)

(c) The MBR shown in the figure below has a reactor volume of V, and it receives a wastewater of flow rate Q. The biomass contained in the influent wastewater is negligible. The effluent from the MBR, i.e., the water passing through the membrane, is nearly free of microbial cells and it has a flow rate of Q_e. Waste sludge is removed from the MBR at a flow rate of Q_w. The biomass concentration (X) in the waste sludge is nearly identical to that inside the MBR. Based on the above information, perform a mass balance for the biomass in the MBR system.

(10 marks)



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